Who Pays the Price of Brokerage? Transferring Constraint through Price Setting in the Staffing Sector

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This article analyzes how a broker's ability to affect prices and extract superior value from its position has economic consequences for the actors tied to it. I argue that intermediaries may exercise partial control in price setting by transferring the price constraints imposed on them by the actors on one side of the market to those on the other side. In so doing, they generate unequal returns for the brokered parties, who then receive different prices due to the nature of the tie between the broker and its other exchange partners. I investigate this argument using a novel mix of quantitative and qualitative data gathered from an intermediary in the staffing sector. The results show that the broker is able to transfer discounts offered to valued buyers (clients) on to the sellers (workers) matched with them, instead of reducing its own margins. As a result, actors with the same resource endowments receive different prices depending on the relationships among other exchange partners in a given triadic network of ties. Using qualitative fieldwork, I elaborate on the mechanisms that make the quantitative results feasible, and I discuss the implications for our understanding of the processes, dynamics, and market stratification consequences of brokerage.

A fundamental characteristic of the sociological perspective on the economy is that the position of actors in the social structure matters to the returns they obtain—over and above their performance or resource endowments (Sorensen 1996). Accordingly, many studies have examined the effects of structural location on actors' economic outcomes. However, it is not often acknowledged that if certain actors can extract superior value from their positions, this might have economic consequences for the other parties tied to them. I investigate this issue in the context of market intermediation.

Among the multiple sources of structural inequality, brokerage has been identified as a key mechanism of market stratification (Burt 1992; Gould and Fernandez 1989). Intermediation both creates value and enables a broker to capture part of this value in the form of rents (see Reagans and Zuckerman 2006). If—as most sociological accounts of brokerage assume—intermediaries extract rents beyond their pure value creation role, we might then expect that someone may be "paying" at least part of the price of rent extraction. So far, though, little is known about whether this is the case and, if it is, what mechanisms make it possible. Addressing this issue highlights a crucial but currently underdeveloped implication of brokerage theory—its consequences for the brokered parties. In fact, while much has been written on the value of brokering as a position, less

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attention has been paid to how intermediaries leverage their ties to other actors and in turn influence the terms of the exchange for them.

One possible reason for the lack of attention to the consequences of intermediation for the brokered parties is that most analyses and predictions within brokerage theory are derived at the ego level of analysis, and their main focus is the benefit that the intermediary obtains from its position. That is, actors who occupy positions of brokerage are compared to those who do not. All else being equal, the former will receive higher rewards than the latter. However, once we shift attention to those who already perform brokerage roles (the case of many market intermediaries), the dyadic ties between the intermediary and each of the other actors become more visible. For example, consider a real estate agent who makes money by matching buyers and sellers of houses. Her commission is typically calculated as a percentage of the selling price, thus she has an incentive for this to be as high as possible (but see Levitt and Syverson 2005)—with obvious consequences for the buyer. But what if the buyer is a major client of the realtor and is constantly looking for properties in the area? Would this affect the selling price and therefore the seller's outcomes? In other words, how do a broker's ties on one side of the market affect the terms of exchange for the actors on the other side of the market? Supplying the answer to this question will uncover some of the price setting mechanisms through which control is exercised in brokered relationships.

This article sheds light on this issue by integrating insights from exchange and power-dependence theories (Cook and Emerson 1978; Cook and Whitmeyer 1992; Emerson 1962) within a brokerage framework. I argue that brokers may be able to exercise control in price setting partly because of their ability to transfer the price constraints imposed on them by the actors on one side of the market to those on the other side. By doing so, they generate unequal outcomes for the brokered parties, who receive different returns not just due to their quality or performance (as a neoclassical economics argument might predict), but due to the nature of the tie between the brokers and their exchange partners on the other side of the market. In other words, if the position of a broker allows rents to be extracted from a given transaction, the position of the brokered parties *vis a vis* the broker influences whether they ultimately "pay" for some of those rents.

The context of my study is the high-skill staffing sector, in which intermediaries match workers (sellers of labor) with client firms (buyers of labor) in return for a proportion of the project fee. The staffing sector is a typical example of a brokered market, and it also happens to be one of the most rapidly growing industries in the U.S. economy according to the Bureau of Labor Statistics. Temporary help services grew by almost 125 percent between 1990 and 2006 (Monthly Labor Review, March 2006). I use a novel mix of field research and statistical analysis of proprietary data to study the mechanisms through which an intermediary in this market manages to exercise control in price setting by leveraging its ties to buyers and sellers. I collected both qualitative and quantitative data from a specialist staffing agency by conducting 48 interviews and assembling a large dataset on workers, clients, and prices. The data include information on a group of 328 highly skilled workers (sellers) who were placed by the staffing firm in 2,091 projects involving 654 clients (buyers) over a five-year period (1998 to 2002). Most of these individuals were assigned to several projects during their time of affiliation with the agency, which allows me to do person fixed-effects analyses.

My empirical strategy consists of estimating the prices that clients pay and workers receive for a given project. I provide evidence of the intermediary's ability to transfer price constraints from buyers to sellers by showing how the discounts offered to more valued clients, rather than reducing the amount of the intermediary's commission, are transferred on to the workers who are assigned to those projects. That is, workers (sellers) who are matched with clients (buyers) that are valuable to the intermediary obtain lower prices, all else being equal. This result is puzzling from the point of view of the standard economic theory of competitive labor markets, which would imply that the same individual placed in the same type of job should be paid the same wage (see Mehta 1998). Yet, I find that the terms of the exchange for workers depend not only on their quality, but also on the set of relationships in which they are located with respect to the broker. That is, the same actor, with the same resource endowments,

obtains different prices depending on the nature of the ties that the broker maintains with other exchange partners. A broker's ability to exercise such discretion in price setting reveals its ability to extract rents beyond the gains of clearing the market by matching buyers and sellers. But most importantly, it highlights the consequences of this ability for the brokered parties (i.e., the sellers/workers), who receive lower prices than would be expected for some of their transactions.

I use insights from my fieldwork to provide an understanding of the circumstances that make the brokered parties "accept" these arrangements. I discard explanations based on individual preferences, and instead I emphasize the importance of social frictions that shelter these broker-seller ties from pure competition. The analysis thus contributes to our knowledge of the ways in which the brokered parties experience brokerage, and it emphasizes social constraint and power dynamics rather than voluntarism and individual choice. I also discuss the external (market-based) and internal (organizational practices) factors that might constrain a broker's ability to exercise full discretion in price setting—and thus pay lower wages across the board. In particular, in the discussion section I address another intriguing puzzle raised by the empirical findings, the fact that the organization seems to work more as a boundedly rational "margin-minder" than as a fully rational "profit-maximizer."

BROKERAGE, CONSTRAINT TRANSFER, AND PRICE SETTING

At the root of a broker's market advantage is the ability to draw information and control benefits from its position, and control is often exercised through price setting mechanisms (see Burt 1992). The intermediary's central location between potential transaction routes allows it to affect the exchange value of the resources it controls (Burt 1977; Marsden 1983). As a result, an actor who brokers between two parties interested in each other's resources will be able to collect a payment for arranging the transaction (Gould and Fernandez 1989; Piskorski and Anand 2004). These payments or "commissions" could be understood as a "fraction of the resource flow between principals" (Marsden 1982:206). The processes by which intermediaries affect market outcomes through their ability to negotiate and exercise control over prices are thus a fundamental underpinning of brokerage theories.

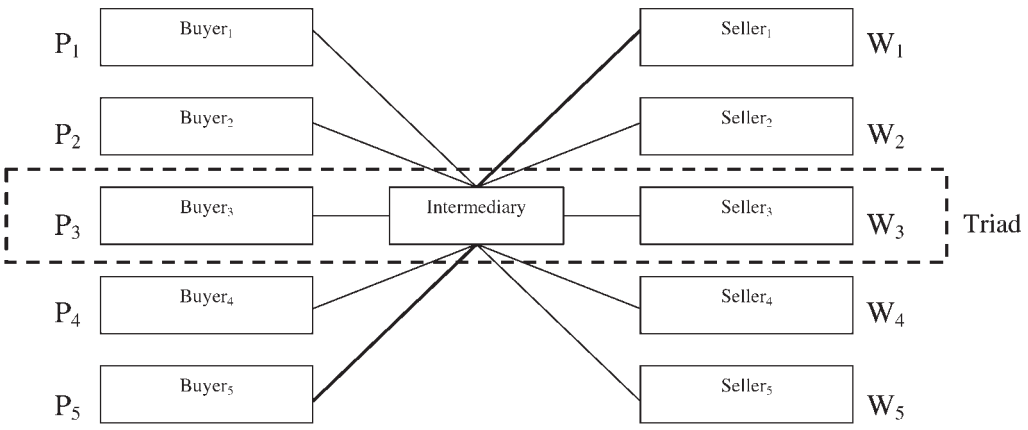
Brokerage and its corresponding "commissions" can potentially take many forms (see Fernandez and Gould [1994] for a theoretical treatment of brokerage structures). In a market context in which intermediaries match buyers with sellers, the broker's commission can be paid by the buyer (e.g., those looking for an apartment), by the seller (e.g., those selling a house), or by neither of them directly. This last is common in labor markets, and it corresponds to the context of my empirical setting (a staffing firm). In such situations, the return to the broker is simultaneously determined by the difference between what the buyer pays and what the seller receives. That is, the company (buyer) who hires the worker (seller) pays a certain amount for the services of the worker to the intermediary. The intermediary, in turn, pays the worker a percentage of this amount and keeps the difference as a commission for its services. This profit margin is the toll that the middleman levies for providing the link between two previously disconnected sets of actors who are interested in transacting with each other (see Reagans and Zuckerman 2006).

The literature on intermediation tends to overlook the tension that arises for the broker when extracting value by setting prices for actors on both sides of the market (who have competing interests in prices). We might expect, however, that the nature of the relationship between the broker and either the buyer or the seller would have some kind of impact on this process. Specifically, relatively higher investments in a given relationship may place greater constraints on behavior. For instance, norms of reciprocity might be stronger (Blau 1964) or power-dependence may be higher (Emerson 1962). These constraints should have a discernible influence on how the intermediary manages to exercise control over the two parties and therefore should affect the terms of the exchange between them. I draw on exchange and power-dependence theories (Cook and Emerson 1978; Cook and Whitmeyer 1992; Emerson 1962), which argue that the conditions of exchange in a given relationship partly depend on the broader set of ties in which the relationship is embedded (see Skvoretz and Willer 1993; Gargiulo

1993). In particular, I claim that brokers will be able to exercise control through their ability to transfer the price constraints imposed on them by actors on one side of the market to those on the other side. This, in turn, has consequences for the prices that the "brokered" actors pay for exchanging through an intermediary.

To see how this "constraint transfer" process would work, imagine an archetypical market intermediary (see Figure 1) and analyze its ties with buyers and sellers. The broker operates by matching sellers to buyers and profiting from the difference between the prices that buyers pay and what sellers receive. This article does not focus on the matching process involved in connecting buyers and sellers but on the price setting process that happens once the matching takes place. I therefore study price setting within a triad that has already been formed. I also set aside issues of competition dynamics in the intermediaries market, although I do assume that the broker is not a monopolist but operates in a sufficiently competitive market. This means that both buyers and sellers have some other options apart from transacting through this particular intermediary. This assumption (which holds in my empirical setting) ensures that a broker's exercise of control in the price setting process is not just the result of its indiscriminate use of market power due to a monopoly position. The theory thus holds constant the number of alternative exchange partners for the actors, in order to focus on the value of the resources transferred through the ties and keep the model in its simplest form (see Emerson [1962] for a discussion of these two sources of dependency in dyadic relationships). In the discussion that follows I start by analyzing the broker's ties with buyers but, in the more general theoretical case, if we started from the other side of the market we could expect to find that the processes would be symmetric.

As has frequently been observed, most organizations have a range of relationships with other actors, from sporadic/arm's-length ties to frequent/repeated transactions (Baker 1990; Kollock 1994). One can reasonably expect, therefore, that an intermediary may have developed a pattern of exchanges over time, in which its level of investment in a relationship is higher with some buyers than with others, and with some sellers than with others (these relationships are represented by thicker lines in Figure 1). Within the framework of exchange theory, ties are conceptualized as the transmission of valued resources—both tangible and intangible (Cook and Whitmeyer 1992). More concentrated exchanges with some parties will make the focal actor (in this case, the broker) relatively more dependent on them for critical resources (Burt 1980; Emerson 1962; Pfeffer and Salancik 1978). As a consequence, when the resources—current and expected—that are exchanged through a relationship are more valuable to the



Pi = Price that the intermediary charges to the buyer
Wi = Price that the intermediary pays to the seller
Broker's Commission = Pi – Wi

Figure 1. Market Intermediation

focal actor, he will have a higher willingness to invest in the relationship long-term, even if it means lowering prices in the short-run (see Kollock 1994; Blau 1964; Emerson 1976; Marsden 1983).

The potential of concentrated exchanges to affect an actor's ability to extract better terms from the transaction is well established—to the point that it is almost taken for granted in many empirical settings. It is less often acknowledged, however, that offering better exchange terms to some clients may be a source of constraint for an organization. In the intermediation context, a decision to charge lower prices to valuable clients imposes (at least in the short-term) a constraint on the broker by threatening to reduce the size of its commission. Moreover, the constraint exists regardless of the reasons why lower prices are offered to these buyers. Dependence and concentrated exchange generate constraint for the broker,[1] but so would norms of reciprocity (Blau 1964), friendship ties (Ingram and Roberts 2000), or embedded relationships (Granovetter 1985; Uzzi 1996). What matters is not the source of favorable treatment for some exchange partners but the fact that such treatment has the potential to constrain the behavior of the actor more so than would arm's-length ties. Sporadic, noncommittal ties might not provide relationship-specific benefits but neither are they binding (see Uzzi [1997] and Portes and Sensenbrenner [1993] for examples of the liabilities of strong ties).

Studies that focus on how relationships between actors affect the terms of the exchange between them are plentiful, but they focus almost exclusively on the economic consequences of these ties for the two related parties. How the nature of dyadic relationships affects the prices obtained by actors who are also tied to those involved in the focal tie, however, has not been fully explored. We do know that a relationship between two actors might be affected in a variety of ways by their ties to third parties (Burt 1999; Coleman 1990; Cook and Whitmeyer 1992). For example, trust or distrust between two individuals is exacerbated by the presence of indirect ties between them (Burt and Knez 1995). Norms of reciprocity and gift exchange in a relationship are also influenced by the actors' ties to others (Bothner, Meadow, and Ozdemir 2006). Yet, the presence of similar effects in price setting processes is not understood as fully. In particular, if relationships are a source of constraint for price setting, the question arises of whether and how these constraints might affect third parties. This is an especially salient issue in mediated markets, as the essence of a broker's advantage is its position as a bridge between two sides of a transaction. A broker who decides to offer lower prices to some buyers faces a constraint (at minimum, a short-term one) because the size of its commission is threatened. However, the commission also depends on the price that the broker pays to the seller when matching it with a buyer—the lower the price the higher the commission. The question is how the intermediary will be able to exercise control in setting prices for sellers in order to manage the constraint created by its offering better prices to buyers.

If the broker has any control in price setting beyond pure matching of supply and demand, one can expect that it will be able to manage this constraint by transferring discounts offered to buyers on to sellers. I call such behavior "constraint transfer," which is used by a broker to reduce dependency by exploiting its tie with a third actor. This idea is closely related to—but slightly different from—Emerson's (1962) "balancing operations." Empirical studies that have analyzed ways of reducing dependency usually focus on the act of establishing a tie with a third party in order to manage constraint (see Gargiulo 1993). In a brokered transaction, however, a tie with a third party is already established but might still be used to manage constraint through price setting. In other words, if prices are a source of constraint, they can also become the way to escape such constraint. If, for a given transaction, a broker charges a lower price to a certain buyer, constraint would be easily avoided if he could pay a similarly lower price to the seller. Of course, this is not so straightforward.

[1] Note that I do not use the term "constraint" in the technical way that Burt (1992:54) and other network researchers do. That is, I focus on the constraint on prices generated within the dyadic relationship between the intermediary and the buyer (due to the value of the resources exchanged), not on the existence of relationships among buyers tied to the broker.

The ability of a broker to transfer constraint imposed by buyers on to sellers requires that the latter accept a lower price for certain transactions. For that to happen, the broker must have the ability to exercise some degree of power over them. After all, if actors pursued their own self-interest, "they would not voluntarily agree to a small share if a larger share were possible" (Skvoretz and Willer 1993:803). For this to happen, there must exist some degree of friction in the market that shelters broker-seller relationships from pure competition. This friction might arise merely from the existence of uncertainty in economic transactions, which creates a certain degree of "stickiness" in exchange relationships. That is, actors operating in conditions of uncertainty about the quality of potential exchange partners will be more likely to stay with their current partners even if they could get a better price from someone else (Kollock 1994). These actors might not even be aware that they could obtain better prices elsewhere, which introduces a further element of "stickiness" in the relationship between the broker and the sellers. If either or both of these conditions hold, one can then understand why sellers would agree to sometimes receive lower prices from a broker. In such cases, a broker can transfer constraint from one side of the transaction to the other.

Therefore, a straightforward test of whether a market intermediary has the ability to exercise control in price setting by transferring constraint will be met if we find that: *sellers who are matched with clients who are more valuable for the intermediary obtain lower prices, all else being equal.*

It is crucial to point out that the step of discount transfer from valuable buyers to the sellers matched with them is not immediately obvious from a purely economic point of view. It would be feasible that if a broker charges lower prices to important clients, instead of transferring these discounts to sellers, it reduces its margins. In such a case, the broker would fully "pay" for the investment in its relationship with more valuable clients. Showing that some of the costs of this investment are transferred to sellers—and not to all sellers, but only those matched with valuable clients—provides evidence that the broker is able to extract value from brokerage beyond its pure matching/value-generation role. More importantly, it highlights a rarely explored implication of brokerage theory: its economic consequences for the brokered parties.

To perform a strict test of this hypothesis, we must show that the matching process is relatively independent of the pricing process. That is, the reason why sellers matched to more valuable clients receive lower prices is not because lower quality sellers are matched with these buyers (which would be consistent with an economic matching argument), but because the broker has the ability to lower the prices for the seller. We must thus show that the *very same seller* receives different prices for the same product depending on the type of relationship that exists between the broker and the buyer. In other words, inequality in economic outcomes for the sellers does not stem from the resources they possess, nor even from their relationships with other actors (the broker), but from the characteristics of the tie between the broker and its buyers, who are one-step removed from sellers in the social structure. This is a distinctively sociological argument about how unequal economic outcomes stem from differential positions and relationships as opposed to mostly differential resource endowments and productivity[2] (Sorensen 1996), and it integrates a basic insight from power-dependence theory into our understanding of brokerage processes. By doing so, it reveals a fundamental—but so far unexplored—implication of brokerage theory: if brokers have the ability to extract rents through their activities, someone may be "paying" at least part of the price of rent extraction. The "constraint transfer" argument specifies one mechanism through which this can actually happen.

I test this proposition using data from the high-skill IT staffing sector. The broker is a temporary help agency, buyers are client firms, and sellers are workers who are matched to these firms. To provide evidence of the broker's ability to transfer constraint, we need to show that the same worker obtains lower prices

[2] More recent developments in economics, such as the matching literature, would contemplate the possibility that similar resource endowments might not be equally rewarded in all positions. The reasons why this would happen, though, are different from those explored in this article, as they mostly focus on productivity differences.

for the same type of job when she is matched with a firm that receives discounts from the broker. Furthermore, we need to provide an explanation for why such highly-skilled workers would knowingly or unknowingly accept lower prices (I do this by drawing on in-depth qualitative fieldwork). An additional robustness check for the main hypothesis explores how ties between the workers and the broker affect the latter's price setting behavior. Further illustrative evidence of the intermediary's exercise of control is provided by showing that the broker does not transfer price discounts uniformly across workers, and also that the broker invests in its relationship with sellers by providing better prices for similar services to those with whom it has more concentrated exchanges.

FIELDWORK: THE STAFFING SECTOR AND THE RESEARCH SETTING

I use a mix of fieldwork and statistical analysis to illustrate the mechanisms described above. In the process of collecting the quantitative data that I use to test my theoretical arguments, I spent 15 months gathering information through interviews and on-site observations, which produced around 325 pages of qualitative field notes. This wealth of contextual, in-depth information serves several purposes that could not be fulfilled by quantitative data alone. First, it provides a better understanding of the specific context of the staffing sector and the functioning of labor market intermediaries in general. Second, it provides a thorough, field-based validation of the quantitative measures. And finally, it helps to clarify the conditions under which the theoretical mechanisms that I propose take place, and it rules out alternative hypotheses for the observed statistical findings. A convergence between quantitative and qualitative methods provides more reliable results (Jick 1979). It also allows for a richer understanding of the observed phenomenon by contributing to the interpretation of statistical relationships and helping to clarify any puzzling findings (Sieber 1973).

The staffing sector in which this research is situated is typical of a brokered market in which intermediaries mediate between buyers and sellers of labor. This industry has experienced steady growth in the last couple of decades (Theodore and Peck 2002), partly fuelled by a broader trend toward the decline of internal labor markets and the rise of contingent employment arrangements (see Kalleberg 2000). Labor market intermediaries allow firms to increase flexibility in their employment relationships by providing them with workers to fill short-term projects (Davis-Blake and Uzzi 1993). The intermediary usually takes care of screening contractors, paying their wages, and performing other administrative functions associated with employment (see Barley and Kunda [2004] for an exhaustive description of staffing firms' activities). Most staffing firms operate by matching workers (sellers) to client companies (buyers) in return for a proportion of the project fee. The client pays the agency a certain sum for each project—usually an hourly rate; the staffing agency in turn pays a proportion of this sum to the worker assigned to the project.

I next discuss the main characteristics of the staffing firm where I did my quantitative data collection, followed by a description of its price setting practices and its relationships with clients (buyers) and workers (sellers).

Research Setting

The research setting is a large, global staffing agency (from now on "InterCo"), headquartered in the United States, that specializes in placing highly-skilled information technology (IT) professionals in a variety of temporary and—occasionally—permanent positions. The focus on highly qualified workers is important, as they will have other potential employment opportunities beyond those provided by the intermediary. For most of the period of my study (1998 to 2002), workers in this sector were in extraordinarily high demand due to the effects of the Internet bubble. This makes any evidence of price control on the part of the staffing firm all the more intriguing and theoretically interesting.

My fieldwork included repeated visits to one of InterCo's offices—located in the same city as its headquarters—during a period of more than a year. I conducted several rounds of interviews with InterCo's managers and observed their day-to-day activities for three to four hours a day, usually three days a week. I also interviewed contractors and several staffing industry experts. In total, I conducted 48 formal

interviews with 43 individuals, in addition to engaging in numerous informal conversations. The interviews lasted for an average of 45 minutes and most were taped and subsequently transcribed. The interviewees include 37 contractors (26 of whom were affiliated with InterCo at some point, although not necessarily at the time of the interview), four of InterCo's managers, and two industry experts.

My preliminary background research on the staffing industry indicates that the practices of this firm do not appear to be systematically different from those of similar specialist "niche" staffing agencies. In the local labor market under analysis there are several firms in competition. According to my interviews, these companies are perceived to be very similar in terms of how they conduct their business. One of the contractors put it as follows: "They are pretty much all the same, [InterCo] probably has the best network of connections in [City], but in terms of the amount of money that they can get, and the amount of commission that they charge, they are all about the same." Some of the workers I spoke to, however, claimed that InterCo was more respectful and attentive to contractors than were other staffing companies. In general, however, I could not discern any major systematic differences between this firm and its immediate competitors.

Workers sign on with InterCo for a variety of reasons. My interviews suggest that many prefer working as contractors rather than as regular employees. Others, however, were contemplating eventually taking a regular position. While my quantitative sample does not include systematic data on why individuals join the agency, there is some information to suggest that an inability to find regular employment is not their main motivation (remember that these are highly-skilled workers in high-demand occupations). Each individual's application to the agency includes a question about whether they are interested in temporary jobs, permanent jobs, or both. Only 1 percent listed permanent employment as their main interest, and 40 percent claimed they would not consider a permanent job as an option. In fact, 35 percent of the workers were contractors at the time of joining InterCo. Whatever the reason why workers decide to work as "temps," using the services of a staffing agency helps their search for projects in client firms: InterCo provides them with assignments and takes care of all the paperwork associated with employment.

InterCo operates by matching workers to projects in client firms. It keeps a database with extensive information on both contractors looking for projects and companies looking for workers. In most cases, it maintains stable relationships with at least some of these clients and contractors. To be included in InterCo's database, workers must pass a selection process consisting of several rounds of interviews and tests, designed to select only those workers that InterCo believes will have a reasonable chance of being placed in the kinds of projects that clients demand. Being included in the database does not guarantee being assigned to a project; and, after a certain length of time, workers who have not been placed are generally removed from InterCo's records. Contractors who are not on assignment do not receive any salary or retainer, nor are they employees of InterCo, and they can be registered with several intermediaries at the same time. When a contractor is placed in a project, it is the staffing firm who becomes the "employer of record" and the worker has no contractual relationship with the client firm. Most assignments have an established duration, which is usually relatively short—a few days or a few months. Some projects, however, are extended. Both clients and workers sign a noncompete agreement with InterCo, which commits them to not contracting directly with each other for a certain period (usually one year) after the intermediary has matched them. These agreements are for the most part honored, and when they are not InterCo does not hesitate to initiate legal action.

The Price Setting Process

The matching begins with InterCo receiving an order from a client and searching its database to find someone who fits the assignment description. This results in a simultaneous matching process in which candidates are offered potential assignments and clients typically are presented with a small selection of resumés to choose from. At the same time, price negotiations begin between InterCo and the client company. InterCo has a recommended range of prices (bill rates) for each type of project, which are detailed in an official document. The specific bill rate within a range is negoti-

ated with each client for each project. Once the bill rate is decided, the chosen contractor is offered a pay rate that is a percentage of the bill rate that InterCo receives from the client. This percentage of the bill rate is not fixed by company policy; it can vary slightly from project to project. Therefore, both bill and pay rates fall within a range of possible values for each category of assignments, but within that range there is a fair degree of variation. It is important to point out that InterCo sets the pay rate for individual workers project by project. That is, it does not make a systematic calculation based on an individual's pay rates in previous projects in order to maintain a consistent wage profile. Workers are free to decide whether or not to accept the job at the rate offered, although both placement agents and workers claim that the majority of the time offers are accepted. Contractors very rarely get to negotiate pay rates in this setting. According to one of the placement agents, "We always negotiate with the clients, not with the worker, we give them a price and they take it or leave it." Workers agree with this view of the wage-setting process:

> You know, I haven't really negotiated with them, they kind of tell you what the rate is.

> I don't really have any control of charging with [InterCo]; they determine my rates.

Individuals who had experience with both InterCo and other similar staffing firms claim that there is usually not much room for negotiation with any of them:

> Generally it is fixed, it seems like when they call they are pretty much giving you an offer, they say, "I have a job for you if you want it, two weeks and this is how much it pays." Rarely, they will renegotiate the rate.

Contractors in this setting are generally unaware of how much InterCo charges the client. InterCo explicitly discourages either clients or workers from discussing prices and wages. This is a key characteristic of most staffing firms, confirmed by previous ethnographic research in the sector (see Barley and Kunda 2004), as well as by my own interviews with contractors. The quotes below are illustrative of these discussions:

> What they can pay you and what they charge the client . . . they don't tell me . . . a little creepy, no one really knows; it is a well-guarded secret. I haven't asked; if I did I don't know if they would tell me.

> [The client] does not talk with me about how much they pay to [InterCo]. I actually don't believe I am really supposed to . . . they probably pay a higher rate to [InterCo], so they can probably, you know, make a profit out of it, which I understand and I am sure all the parties included understand. I don't really discuss with them how much I am receiving from [InterCo] either.

InterCo's Relationships with Buyers and Sellers

InterCo's buyers are the companies that hire contractors for short-term employment projects. Some of these firms do business with InterCo very infrequently, hiring workers only occasionally. Other firms, however, generate a significant part of InterCo's business and are considered by its managers to be "best clients." According to one of InterCo's managers, "My main definition of best clients is those who bring a lot of business in." Valued clients, therefore, are those that provide high volumes of exchange, but this is not the only criterion that InterCo uses to decide whom its best clients are. The company keeps a list of "preferred clients," which includes those firms that are considered to be most valuable to them. There is no official companywide policy about which clients should be included in the "preferred lists," which differ depending on the local market and are mostly under the control of each regional manager. When probed about the decision-making process by which firms are included in the "preferred list," InterCo's managers claimed, "How we come up with the list is not put in writing anywhere, but it's generally how we all kind of talk about it." Interviews with InterCo's employees indicate that companies included in the list fulfill several criteria. First, they generate a large and stable volume of business for InterCo, often hiring large numbers of contractors. Second, they offer InterCo the potential to grow its business, as they are generally large companies operating in industries with multiple departments that require highly specialized IT contractors. According to the managers:

> The size of the company is very important, meaning that there are lots of different departments and divisions within the company that we can work with, and so there are lots of opportunities. . . . If they are large companies but they have their own people in there that know that side of the business

> and they rarely ever need a large amount of people placed, they wouldn't be considered a preferred client.

> It is usually companies that are headquartered in [City], because it is in headquarters where there is more of a potential for us.

> It really depends on the potential of the relationship, whether we have a contact at the high level in headquarters (VP, etc.). . . . The potential that we see in that client to expand our piece of the pie.

The potential of the relationship (expected value of the resources exchanged) is therefore the second key ingredient of a "valued" tie, which goes beyond—but is correlated with—current and past volume of business. As one manager said, "Some companies can be high volume and high sales, but they wouldn't be 'preferred' because there is no place else to go." The potential of the relationship is also related to InterCo's knowledge of those companies: "It is important that we understand their business very well, which needs they have," as this knowledge allows for a clearer growth strategy within those client companies. In fact, as one of the placement agents put it, "The regional managers want to see movement on the preferred list; that you have tried to get in every part of the business that you can with that company." Current and potential volume of exchange is therefore the best predictor of a "valued relationship" in this setting. This definition corresponds to Cook and Emerson's (1978) understanding of "committed ties" as those through which both tangible and intangible resources are exchanged. In this context, it is also a definition of InterCo's perception of the client's strategic importance.

The "preferred clients" list is fairly stable, and it did not change during the period of this study. In fact, managers claimed that relationships with these buyers are usually long-term: "It is about the long-term character of the relationship or about the depth of the relationship." They also claimed that repeated exchanges with these companies sometimes led to closer personal relationships with the hiring managers: "If you are doing your job well, you need to visit, you need to have conversations with them, and it just happens in the course of business . . . you sometimes develop a more personal relationship." Moreover, this development of personal ties with some hiring managers may increase business from those companies:

> Doing business with these companies is about maintaining the relationships with the people who do the hiring, the hiring managers; finding out from these people who make the decisions. They help me make new contacts to expand the business.

The general perception among InterCo's managers then, is that the best clients are also the most crucial for their business and those on whom they are relatively more dependent. Accordingly, they acknowledge that they are often willing to offer better terms in their transactions with them:

> We cater to clients we have established relationships with, and we do anything to keep them happy . . . larger clients sometimes get discounts; it is common, although we don't like to call them volume discounts.

> We might be inclined to give them a better rate if they order from us on a consistent basis.

My fieldwork, therefore, suggests that InterCo has a number of clients who are especially valued, and there is at least anecdotal evidence that they are willing to offer these clients better prices. The main hypothesis in this article, however, is that the intermediary will be able to transfer these discounts to the other side of the market—the workers who are matched with the valued clients. InterCo has no official—or even unofficial—stated policy for doing this. The only way to test this hypothesis is by using actual price and wage data, which I analyze in the next section.

During the course of my fieldwork, I also investigated the nature of the relationships that InterCo maintains with the other side of the market (the workers). The interviews with contractors and managers indicate that, similar to what happens with clients, InterCo values relationships with some workers more than with others. In this case, however, there is no "preferred worker" list. Instead, both managers and contractors talked about how workers who had spent longer periods of time with InterCo tend to develop a more "committed" relationship.[3] When asked to identify who their more valued

[3] I use the term "commitment" in the sense of Cook and Emerson (1978), to indicate behavioral patterns of exchange. This article does not make any assumptions or claims about the emotional commitment of these actors.

contractors are, placement agents were in agreement that they are "those who have been with us for longer and therefore have been placed in a lot of projects." There seems to be a high correlation between the volume of exchange generated by contractors and the duration of their relationship with InterCo. This is because those who are not placed regularly are dropped from the database (while clients, even if they generate very little business, are rarely dropped from the database). Contractors who repeatedly work with InterCo tend to be valued more highly by placement agents, partly because they know they can place them almost anywhere and they will do a good job. These individuals are also appreciated because often they bring more business in, usually by giving InterCo information about job openings or names of contacts in client firms. Over time, and as happens with clients, placement agents also develop a more personal relationship with these workers. For example, one of the managers said: "We treat them differently in the sense that it is more casual, it's not like there is preferential treatment, but when we talk on the phone it is the difference between talking with a friend and talking with someone you have just met." These relationships are complex and include elements of both repeated exchange and increasing personal involvement over time. The end result is that certain exchange partners are considered by the broker to be more valuable.

As is the case with preferred clients, InterCo acknowledges that it is normally willing to offer better prices to these workers. One of the managers explained: "If there are two comparable people and the same assignment, we may pay more to the one who has been working with us for longer." Some workers (although not all) seem to understand that this happens, and they believed their negotiation power on wages would increase if they kept bringing in more business for InterCo. For example:

> [InterCo] has a pay scale, but they have flexibility, a couple of dollars an hour to negotiate, they may pay me a couple more dollars an hour because they know me, they know I will do a good job in the client.

> I had the sense that if I stayed working with them for a longer period of time they would increase my rates. I don't know if it was a realistic sense, but that was the impression I had.

As the description of this intermediary shows, the importance of relationships on both sides of the market seems to be perceived as having a significant influence on the prices that the broker charges to buyers and pays to sellers. In the next section, I describe the quantitative data used to test the proposition that the broker is able to leverage these ties to transfer constraints generated on one side of the market (clients) to the other side of the market (workers).

QUANTITATIVE ANALYSIS: SAMPLE AND METHODS

I test my arguments using InterCo's proprietary information on contractors' job histories and client data. To focus the study on a single geographic and occupational labor market—and therefore limit potential sources of unobserved heterogeneity—I restrict the analysis to one of the company's subsidiaries, located in the same city as the headquarters. I assembled the dataset from a variety of sources provided by InterCo itself (paper resumés, client information, demographic data, project characteristics, prices, etc.), as well as public information on size and industry classification of clients. I follow the trajectories of the 328 individuals that joined InterCo in 1998 and 1999 and were placed in temporary assignments, from their starting date as entered in the database to September 2002. This group of workers was placed (most of them repeatedly) in 2,091 projects across 654 different companies during the period.

My empirical strategy consists of estimating the prices that clients pay and workers receive for a given project.[4] If the constraint transfer mechanism operates in this setting, we would expect that more valued clients pay lower prices and workers who are matched with them receive lower wages (controlling for every other variable expected to affect prices and wages). Note that wages paid by InterCo must be within a range determined by the external market (which here is held constant). If the parties could get

[4] This database allows us to analyze the effect of relationships on prices on individual-by-firm transactions. This level of detail is unique in the literature. Uzzi and Lancaster (2004), for example, use average firm prices as opposed to the specific prices charged to each client.

much better prices elsewhere, they would not transact with this particular firm (see Marsden 1983). Wages, however, are not fully determined by competitive forces external to the firm. Labor markets always leave a "zone of indeterminacy" (Rees 1993) within which there is room for bargaining and discretion. Wages cannot go below a certain lower limit—which is unobserved; but they can vary within the "zone of indeterminacy." In this article, I analyze the variables that determine exactly how much InterCo pays within that acceptable range.

Price and wage regressions were estimated using person fixed-effects models, which analyze within-person variation rather than differences across individuals. The use of fixed effects is essential for testing the hypothesis, which requires that *the same* worker receives lower wages when matched with a client highly valued by the broker. If we do not fully control for individual quality, finding that wages are lower for projects with certain clients could simply be the result of matching processes rather than pricing processes. That is, it could be that these clients simply get lower quality workers for the lower prices they pay (which would be consistent with a purely economic argument). Because this article aims to evaluate whether actors with similar endowments obtain differential returns due to their positions with respect to the broker, the analysis needs to fully control for "quality." Running wage regressions by individual ensures this.

Dependent Variables and Controls

The key dependent variables in this article are the hourly bill rates that clients (buyers) pay to InterCo for a given project and the hourly pay rates that workers (sellers) receive from InterCo. In the dataset, bill rates are between $19.5/hr and $150/hr (the average is $45/hr) and pay rates are between $13/hr and $100/hr (the average is $26.98/hr). Both variables are logged.

Individual characteristics. The sample is slightly skewed toward males (53 percent of the individuals), with ages ranging from 18 to 56 years old. Most of these contractors (79 percent) have at least a bachelor's degree. Relevant previous experience variables were coded from the resumés. In separate analyses I checked for the possibility of self-selection bias in the panel due to individuals leaving the sample over time. My results indicate that selection bias is not a major issue, since stayers and leavers do not differ in any key observable characteristics. Moreover, a logit model of the probability of being affiliated with InterCo in September 2002 confirms that none of the individual level variables predict the probability that a worker will be affiliated with InterCo at the end of the window of observation (results available from the author). Note that, because I am using individual fixed-effects regressions, the coefficients for person-specific characteristics do not vary across projects and are therefore not estimated in the models. Also, the regressions are only estimated for those contractors who are placed in more than one project (80.2 percent of individuals in the sample, and 96.89 percent of the sample of projects).

Project characteristics.

- Type of project: The 2,091 projects are classified into eight different skill categories: software, Internet, copy, administrative, database, design, graphics, and communications. Internet and graphics positions are the largest categories in the sample, accounting for 53 percent of the projects. Interviews with placement agents confirm that these classifications are a good measure of the skills required for a particular assignment. InterCo invests a great deal of effort in deciding these classifications, which appear in brochures and documents distributed throughout the company.
- Duration of the assignment: Projects range from 0 days (just a few hours) to 1,070 days, with the average duration being 40 days.

Client characteristics.

- Industry classification: Using information from both InterCo's database and a variety of external sources, including firm Web sites and directories, such as Hoovers and CorpTech, each of the 654 clients is classified into one of 13 broad industry sectors: advertising, architecture and engineering, nonprofit, computers, retail, financial services, healthcare, professional services, pharmaceuticals, media, publishing, telecommunications, or a residual category for firms whose industry was not clearly identifiable.
- Client size: This is measured by "number of employees." I was able to code size for 578 out of the 654 clients (1,915 out of 2,091 projects). Size ranges from 1 to 370,000 employees.

Measures of Relationships with Buyers (Clients) and Sellers (Workers)

A key methodological issue in this article is identifying measures of relationship importance that capture the findings from the qualitative fieldwork while being theoretically meaningful and parsimonious—such that they can be used in statistical models (see Uzzi 1999). The literature on interorganizational relationships uses a variety of empirical measures to operationalize the level of "commitment" (Cook and Emerson 1978) in buyer-seller relationships. Studies in the resource dependence tradition tend to use the volume of exchange between two actors as an indicator of dependence (see Finkelstein 1997; Mizruchi and Koenig 1986; Pfeffer 1972). Scholars who use an embeddedness framework generally consider relationship duration to be a good indicator of embedded ties (Seabright, Levinthal, and Fichman 1992; Uzzi and Lancaster 2004), but measures of concentrated exchange have also been used for this purpose (Uzzi 1996, 1999). Ultimately, disentangling what observed patterns of concentrated exchange actually mean requires a deep understanding of the specific empirical setting. For example, Uzzi (1999) uses fieldwork findings to justify why concentrated exchange is not an indicator of resource dependence in the small business lending sector. In my research setting, however, the importance to InterCo of large clients' business is quite significant.

I have argued that a higher volume of exchange at minimum generates a certain level of dependence from key clients. I remain agnostic, however, as to whether some of the mechanisms invoked by the embeddeddness framework (fine-grained information transfer, joint problem-solving, etc.) are also at play here, as they were not directly observable in this setting. The interviewees emphasized the importance of current and potential sales volume as a source of committed exchanges; they also suggested the existence of a certain level of dependence on key clients and indicated that sometimes they develop personal relationships with them. When asked how valued clients could be identified without asking the placement agents directly (see Uzzi [1996, 1999] for a similar empirical strategy), they were unanimous that one should pick the companies listed in InterCo's database as "preferred." The "preferred" label incorporates an indicator of "relationship potential" that is absent in measures of "past sales volume." In order to keep the quantitative measures as close as possible to the processes observed during the fieldwork, I use InterCo's classification to capture relationship importance, generating a dummy variable that takes the value "1" if the client belongs to the "preferred" list. While only 3.2 percent of the 654 clients are classified as "preferred," they account for 12.14 percent of the projects in my sample.

Besides "preferred client," I use two other measures of client-agency relationships to check the robustness of the results to different specifications of concentrated exchange: "sales volume" and "number of contractors employed at the client." Sales volume measures the total income that a given company has generated for InterCo over the period studied and for the projects being analyzed. It ranges from $208 to $2.2 million, with the median being $59,900. I also construct a variable that measures how many other workers within this sample are employed by the client at the starting point of each project. Its value varies from 0 to 15, with an average of .89. These two variables are very unevenly distributed, and one can expect that their effect might not be incremental as volume rises but might be stronger at very high values. To account for this, in the regression models below I created a dummy that identifies those clients who are in the 10th percentile of sales and "number of people." Both measures are highly correlated with "preferred client": 62.2 percent of projects in the "preferred" category overlap with "large sample sales," and 60 percent of those within the "large number of contractors" category overlap with "preferred." These two measures of client importance are imperfect, because they are captured only for the sample of workers and projects that I am studying—not for InterCo as a whole. However, they represent a reasonable proxy for the general patterns of InterCo's transactions, as there is no reason to believe that the distribution of nonobserved projects is significantly different from that of the observed ones. Managers at InterCo, however, believe that the "preferred client" list is a better measure of relationship importance because of its attention to the potential of the tie. They also argue that this classification has a greater influence on decisions in

InterCo than do sales volume or number of contractors employed.[5]

For comparability with other studies of interorganizational relationships, I also calculate the duration of the tie between the client and InterCo. This is measured as the length of the client's relationship with InterCo (number of days) at the beginning of a given project. It ranges from 0 (just a few hours) to 4,915 days (approximately 13 years). As mentioned above, this is not a good indicator of relationship importance in this setting, because clients are rarely dropped from the database. Therefore, duration does not adequately capture concentrated exchanges or repeated transactions with buyers.

On the workers' (sellers) side of the market, there is no comparable measure to the "preferred client" list. Based on the insights from my fieldwork, I use the duration of the tie between the worker and InterCo ("tenure") as a measure of relationship importance. This counts the number of days that an individual has been in the database at the time of being placed in a particular project. It ranges from 0 to 1,692 days.

For comparability with at least one of the client tie variables (sales volume), I calculate the amount of money generated by each individual at the starting point of each project, which I call "bills generated." I multiply the number of days that a worker was on assignment in the past by the bill rate for those projects (assuming eight hours per day). This variable is highly correlated with "tenure" (.64, $p = .000$). However, it disproportionately favors contractors placed in the most skilled positions (e.g., Java programmers with bill rates around $130/hr). Table 1 contains summary statistics for the key variables (for a complete correlation matrix, please see the Online Supplement on the *ASR* Web site: http://www2.asanet.org/journals/asr/2007/toc056.html).

RESULTS

Table 2 presents estimates of project bill rates. Model 1 indicates that clients on the "preferred" list obtain lower prices from the broker for the same type of project and for the same worker. Notice that firm size by itself has a positive effect on bill rates, which means that larger companies pay higher prices in general. However, those clients that InterCo considers more "valuable" pay lower prices —even though for the most part these are large companies. All models include controls for the project's skill requirements and for the client's industry affiliation. The regressions also include "year in which the project began" to control for possible variations in the economic cycle, worker's tenure with InterCo (logged because this variable is highly skewed), and project duration.

Models 2 and 3 in Table 2 use the two additional indicators of concentrated exchange. Model 2 introduces a dummy that takes the value "1" for those companies in the top 10 percent of sales with InterCo. This coefficient is also negative and highly significant on bill rates. Similarly, companies in which InterCo has more contractors at the beginning of a given project also pay lower bill rates (see Model 3). If we include all three independent variables in the same model, their significance decreases because they are all highly correlated and are proxies for the same theoretical construct. An F-test, however, shows that they are jointly significant (results available from the author). Model 4 indicates that, as expected, the duration of the relationship between the client and InterCo has no effect on prices.

Notice that "preferred clients" are not defined by InterCo as "those who get discounts"; that is, receiving lower prices is not an intrinsic characteristic of this independent variable. First, the list of preferred clients is established separately from and previous to the price setting process. Second, as mentioned above, prices are set for each individual project, and they fall within a range of accepted prices for each type of skill requirement. Finally, while projects with preferred clients on average have lower bill rates than projects with nonpreferred clients, not all do so. There is a certain amount of overlap in the distributions of bill rates for both types of clients, which suggests that receiving a discount is not a necessary characteristic for being a preferred client.[6]

[5] To explore quantitatively the variables that influence being a "preferred client," I implemented a logit model to predict this variable as a function of a series of client characteristics. For more information on this issue, see the Online Supplement on the *ASR* Web site.

[6] Similarly, there is a significant amount of variance in the size of the commission (bill rate – pay rate)

Table 1. Summary of Descriptive Statistics

Variable	Mean	SD
Ln Pay Rate	3.22	.36
Ln Bill Rate	3.75	.34
Preferred Client	.12	.33
Ln Sales Volume	10.85	2.12
Sales Volume Top 10th Percent	.09	.28
Number Contractors at Client	.89	1.79
Number Contractors Top 10th Percent	.07	.26
Duration Client at Assignment	1,514.77	1,328.47
Tenure at Assignment	321.74	354.62
Ln Bills Generated	10.20	1.67
Project Duration	39.96	78.85
Ln Client Size	6.11	2.87
Industry Missing	.01	.09
Industry Advertising	.21	.41
Industry Architecture and Engineering	.01	.10
Industry Nonprofit	.08	.27
Industry Computers	.12	.32
Industry Retail	.12	.33
Industry Financial Services	.19	.39
Industry Healthcare	.01	.08
Industry Professional Services	.11	.31
Industry Pharmaceuticals	.05	.22
Industry Media	.03	.17
Industry Other	.00	.05
Industry Publishing	.05	.21
Industry Telecommunications	.01	.12
Project Software	.07	.26
Project Internet	.25	.43
Project Copy	.02	.13
Project Administrative	.02	.14
Project Database	.03	.17
Project Design	.17	.37
Project Graphics	.28	.45
Project Communications	.16	.37
Gender (1 = Female)	.48	.50
Years Education	15.84	1.18
College in Field	.52	.50
Special Training	.30	.46
Years Experience	6.85	4.83
Field Experience	.90	.31
Contractor Last Job	.39	.49

Note: Summary statistics are presented at the project level.

The results in Table 2 are just the starting point to test this article's proposition about constraint transfer. I claim that—for whatever reasons discounts are offered to best clients—the intermediary will be able to pay lower prices to workers when they are matched with one of these companies. Table 3 shows the results of the wage analysis.

Model 1 in Table 3 indicates that, as expected, workers who are matched with preferred clients receive lower wages, everything else being equal. Indeed, the model indicates that *the same worker* will receive a different wage for

that the broker receives for each type of client. Margins average 69 percent for both preferred and nonpreferred clients, with a standard deviation of .15 for the former and .16 for the latter. InterCo does not apply a uniform commission structure; that is, preferred clients are not by definition those for whom commissions are lower. See the Online Supplement on the *ASR* Web site for additional analyses on this issue.

Table 2. Bill Rate Regressions, Individual Fixed-Effects Estimation

	Model 1 Ln (bill rate)	Model 2 Ln (bill rate)	Model 3 Ln (bill rate)	Model 4 Ln (bill rate)
Preferred Client	–.044256**	—	—	—
	(.017078)			
Large Sales Volume Client (10th top percentile sales)	—	–.112705**	—	—
		(.020229)		
Large Number of Contractors at Client (10th top percentile)	—	—	–.087456**	—
			(.018206)	
Duration Client Tie	—	—	—	–.000004
				(.000004)
Ln (client size)	.003971*	.004120*	.003391	.003957
	(.001919)	(.001888)	(.001890)	(.002026)
Ln (worker's tenure)	.016634**	.017037**	.017449**	.016289**
	(.005067)	(.005009)	(.005032)	(.005082)
Project Duration	–.000081	–.000013	–.000035	–.000105
	(.000068)	(.000069)	(.000069)	(.000068)
Year_1999	.093826**	.099450**	.086806**	.090531**
	(.018220)	(.018052)	(.018092)	(.018299)
Year_2000	.177037**	.184570**	.162975**	.174999**
	(.024673)	(.024443)	(.024633)	(.024765)
Year_2001	.229415**	.239241**	.213296**	.229593**
	(.029276)	(.029001)	(.029252)	(.029357)
Year_2002	.265190**	.275466**	.253983**	.269775**
	(.034757)	(.034376)	(.034602)	(.034888)
Type of Project Controls (8 dummies)	Included	Included	Included	Included
Industry Controls (13 dummies)	Included	Included	Included	Included
Constant	3.561733**	3.546576**	3.568857**	3.575839**
	(.112233)	(.111003)	(.111313)	(.112539)
Observations	1328	1328	1328	1328
Number of Individuals	235	235	235	235
R-squared (within)	.33	.35	.34	.33

Note: Standard errors in parentheses. * $p < .05$; ** $p < .01$ (two-tailed tests).

the same type of work (skill requirements, industry, project duration, etc.) if the client happens to have a special relationship with InterCo. Lower wages are not the result of lower quality workers being matched with preferred clients (which would be a matching mechanism), but they result from the broker's ability to lower the worker's price.[7] To see how this works, let us imagine that a programmer was placed in a project last month with a "nonpreferred" client and was paid $100/hr. Today she is assigned to a "preferred client" to work in the same type of project, in the same industry, in a similar sized company, and she receives $93.4/hr. Keep in mind that this is the same person performing the same type of work; it is not the case that a lower quality individual is being placed in the lower paying project and that is the reason for the lower pay. The magnitude of the wage decrease is not huge, but it is economically meaningful. For a worker at the average point of the wage distribution ($26.98/hr), the difference would amount to $1.78/hr (around $70/week). When

[7] One might think that, even if the quality of a worker is exactly the same regardless of the type of client in which she is placed, her motivation to work in projects that pay a bit less might be lower, and therefore her performance would suffer and preferred clients would get a "worse deal." Although it is hard to imagine why the intermediary would knowingly risk underserving preferred clients, this is theoretically plausible. To rule out this possibility, I performed a set of additional analyses (see the Online Supplement on the *ASR* Web site).

Table 3. Pay Rate Regressions, Individual Fixed-Effects Estimation

	Model 1 Ln (pay rate)	Model 2 Ln (pay rate)	Model 3 Ln (pay rate)	Model 4 Ln (pay rate)
Preferred Client	–.063979** (.0149644)*	—	—	—
Large Sales Volume Client (10th top percentile sales)	—	–.099292** (.017821)	—	—
Large Number of Contractors at Client (10th top percentile)	—	—	–.033572* (.016179)	—
Duration Client Tie	—	—	—	–.000000 (.000004)
Ln (client size)	.004200* (.0016812)	.003993* (.001663)	.003343* (.001680)	.003392 (.001785)
Ln (worker's tenure)	.020470** (.0044397)	.020512** (.004412)	.020193** (.004472)	.019688** (.004479)
Project Duration	.000090 (.0000599)	.000135* (.000061)	.000080 (.000061)	.000052 (.000060)
Year_1999	.060338** (.0159653)	.064075** (.015903)	.055352** (.016078)	.057110** (.016127)
Year_2000	.144268** (.0216197)	.150314** (.021533)	.137758** (.021890)	.142659** (.021825)
Year_2001	.195750** (.0256525)	.204503** (.025548)	.189745** (.025995)	.195999** (.025872)
Year_2002	.244251** (.0304556)	.254965** (.030284)	.243072** (.030749)	.248661** (.030745)
Type of Project Controls (8 dummies)	Included	Included	Included	Included
Industry Controls (13 dummies)	Included	Included	Included	Included
Constant	2.937427** (.0983429)	2.999637** (.097788)	3.021046** (.098919)	2.955507** (.083216)
Observations	1328	1328	1328	1328
Number of Individuals	235	235	235	235
R-squared (within)	.37	.38	.36	.36

Note: Standard errors in parentheses. * $p < .05$; ** $p < .01$ (two-tailed tests).

asked whether they would be concerned about wage differences of this magnitude, most contractors claimed that they would. The magnitude of the wage cuts that workers suffer when assigned to important clients would be even bigger if we calculated them using the coefficient on "large sales volume" (10.4 percent), but they would be smaller if we use "number of contractors" (3.41 percent).

Examining how broker-worker ties affect prices provides an additional robustness analysis for the argument that the broker is able to exercise control in pay setting by leveraging relationships on both sides of the market. That is, one might expect that InterCo will pay higher prices to more valued contractors and, in turn, charge higher prices to the average client. The results in Tables 2 and 3 suggest that this does in fact occur. As argued above, the "tenure" variable in this setting is the best available proxy for a "committed" broker-worker tie. As the results show, workers who have been affiliated with InterCo for longer periods of time receive higher pay rates (see Table 3), and this also translates into higher bill rates for InterCo (see Table 2). As Table 4 (Models 1 and 2) indicates, this result also holds if we use the "bills generated" variable as a measure of worker relationship importance.

Finally, further evidence of the broker's control in price setting is shown in Model 3. If InterCo uses its relationships with workers to transfer constraints generated by clients, one might argue that those workers who are also valuable to InterCo would take less of a hit in their wages when matched with preferred clients. Model 3 introduces a dummy variable that takes the value "1" for those workers who

Table 4. Worker-Broker Ties

	Model 1 Ln (pay rate)	Model 2 Ln (bill rate)	Model 3 Ln (pay rate)	Model 4 Ln (bill rate)
Preferred Client	–.063134**	–.055351**	–.068593**	–.047013**
	(.016785)	(.019056)	(.015124)	(.017283)
Ln (bills generated by worker)	.013566**	.014559**	—	—
	(.004590)	(.005211)		
Ln (worker's tenure)	—	—	.020531**	.016497**
			(.004450)	(.005086)
High Tenure Worker × Preferred Client	—	—	.127151*	.076521
			(.063188)	(.072211)
High Tenure Worker	—	—	–.001002	–.014698
			(.029585)	(.033810)
Ln (client size)	.004436*	.004699*	.004176*	.003896*
	(.001852)	(.002102)	(.001685)	(.001925)
Project Duration	.000100	–.000052	.000077	–.000086
	(.000065)	(.000074)	(.000060)	(.000069)
Year_1999	.054155**	.084258**	.059985**	.094505**
	(.017555)	(.019931)	(.016057)	(.018350)
Year_2000	.153020**	.177510**	.143009**	.177777**
	(.022459)	(.025499)	(.021823)	(.024939)
Year_2001	.203964**	.227807**	.193361**	.231363**
	(.026742)	(.030361)	(.026578)	(.030374)
Year_2002	.267995**	.281598**	.234531**	.267456**
	(.030620)	(.034764)	(.034901)	(.039884)
Type of Project Controls (8 dummies)	Included	Included	Included	Included
Industry Controls (13 dummies)	Included	Included	Included	Included
Constant	3.095006**	3.664743**	3.008293**	3.562457**
	(.137329)	(.155917)	(.098248)	(.112277)
Observations	1132	1132	1328	1328
Number of Individuals	198	198	235	235
R-squared (within)	.34	.33	.37	.33

Note: Standard errors in parentheses. * $p < .05$; ** $p < .01$ (two-tailed tests).

are at the top of the tenure distribution (highest 5th percentile)[8] and an interaction term between this dummy and "preferred client." The results show that the interaction term is positive and significant, which suggests that, in contrast to the average worker, InterCo does not "penalize" individuals with whom it has committed ties in order to favor preferred clients. Model 4 indicates that preferred clients, however, do not pay more for high tenure workers (the interaction term is not significant on bill rates), which makes sense if InterCo is still investing in its relationships with them. Because of the small number of individuals captured in these tests, these additional results are not as statistically powerful as the main test of my proposition. Yet, they provide further illustrative evidence of how the "constraint transfer" mechanism works.

Why Do Workers Accept It?

The final step in the analysis requires an understanding of why sellers (highly skilled workers) would accept an arrangement in which they get lower prices when matched with the broker's

[8] I use this dummy variable instead of an interaction ln(tenure) x "preferred client" because the implication of my theoretical argument would only apply to those workers who are at the very top of the distribution, not uniformly as tenure increases. The interaction term with ln(tenure) is also positive but not significant. Different cutoffs of the tenure distribution are also positive when interacted with "preferred client," but only significant for workers with the longest affiliation with InterCo (top 5 percent tenure).

preferred clients. This should help to clarify the mechanisms underlying the statistical results and contribute to our understanding of how the "brokered parties" actually experience brokerage. As argued above, for this to be possible there must be some kind of friction that shelters these broker-seller relationships from pure competition. Part of this friction is simply created by the fact that, for the most part, workers are not involved in bill and pay rate negotiations. My fieldwork suggests, however, that there are other important reasons why InterCo's exercise of control in price setting over workers is possible. First, most contractors simply do not perceive that they receive lower wages when matched with preferred clients (note also that the workers do not know who the preferred clients are). They do know that the level of their wages varies for very similar projects. The qualitative evidence suggests, however, that they do not perceive these wage variations to be systematic or in any way related to the client-agency tie. They usually assume the variations are related to the client's budget for that particular project. For example, when asked where variations in wages come from, typical responses are:

> You may work one week and then the next week the pay may be better, the pay may be worse . . . I think it depends on what the company can budget for freelance.

> I've been fortunate to pretty much demand steady amounts for different types of work; there has been only one or two cases, you know, where a certain client could only afford one or two dollars an hour less.

Most of these contractors perceive themselves as operating in a system relatively close to a pure market, and certainly they do not appear to feel that InterCo is in any way making arbitrary decisions on rates: "[InterCo] is very knowledgeable about what the going rates are . . . they have never asked me to do something for an unreasonable low rate."

Wage distributions for preferred versus nonpreferred companies overlap considerably, which makes it difficult for workers to see that pay variations map onto certain categories of clients. As Figure 2 shows, not all preferred projects pay more than all nonpreferred ones, and vice versa. Consequently, most contractors do not perceive that, on average, they get systematically lower pay rates when assigned to certain companies.

The broker also manages to keep one side of the market isolated from the other by making

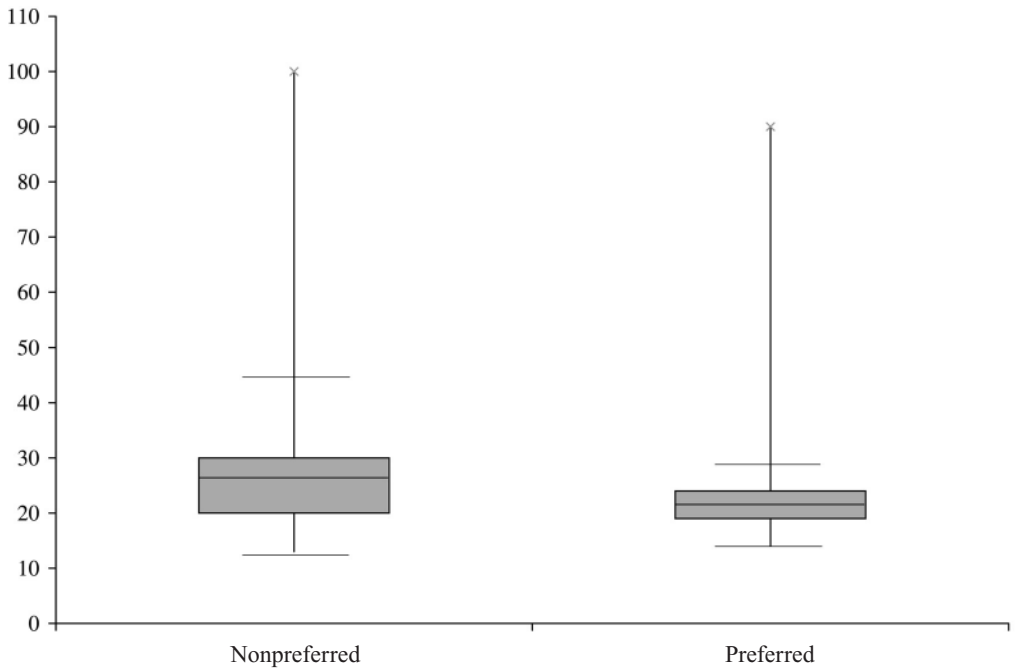


Figure 2. Pay Rates in Nonpreferred Versus Preferred Clients

sure that workers do not know how much InterCo is charging for their services. As one of my interviewees put it, "[InterCo] keeps it so that the right hand doesn't really know what the left hand is doing, as far as the billing; they want to keep their contractors very insulated from the billing process." More importantly, workers have little knowledge of the type of relationship that exists between InterCo and its clients. When I probed this issue, typical answers were:

> I am not aware of that, no.

> They will often say this is one of our best clients, it is very important, or it is a new client and we want to impress them. It helps us to know that to do the work . . . I have no idea of how that might affect rates . . . we are not told about negotiations with the client.

Besides the opaqueness of the mechanisms that InterCo uses to set prices on both sides of the market, a further element that creates stickiness in these relationships is that, over time, individuals get used to working with InterCo and prefer to continue transacting with them rather than facing the uncertainty of the prices they might be offered elsewhere (see Kollock [1994] for a similar argument). As one contractor put it, "It is hard to predict what your pay rate is going to be, as it changes from job to job."

My explanation for why workers "accept" lower wages when matched to InterCo's best clients emphasizes social constraint, not voluntarism. A straightforward economic explanation for the same finding, however, would focus instead on the role of workers' preferences. That is, one could argue that, even if workers do not know which clients are on the "preferred list," they might still be willing to accept lower prices when working for these companies. This would happen if there were something about these clients that made them more attractive for contractors. For example, workers might anticipate that accepting these assignments would increase their future probability of being placed in temporary or permanent positions.[9] If this were the case, they might be willing to sacrifice current wages in the expectation of future benefits, which would be consistent with an interpretation of the results in Table 4 as the realization of "career concerns" (see Gibbons and Murphy [1992] for a review of career concerns issues). It would then not be the case that the broker exercises discretion to lower wages across clients, but that the workers' utility is different for preferred projects—and therefore wages are also different.

To address this preference-based explanation, I evaluate whether working for InterCo's most valued clients improves the workers' chances of either working more in the future or getting a permanent job with the client (see Table 5). Model 1 uses OLS regression with robust errors by individual to predict who will work the longest for InterCo at a particular point in time. The dependent variable is "future days worked." The results show that working for a preferred client has no effect on how much the contractor ends up working through InterCo in the future. Model 2 replicates these results by applying the other two measures of client importance—sales and "number of contractors." The findings are the same (and these variables are not significant if entered separately in the models).

Models 3 and 4 in Table 5 estimate a logit regression—with robust errors by individual—using "permanent placement" as the dependent variable. In this sample, 32 of the contractors ultimately took regular jobs with one of these companies. However, the type of client in which they work as contractors does not predict their probability of placement. This evidence suggests, therefore, that individuals' preferences for certain types of clients do not seem to be driving the discount transfer results.[10] These findings were confirmed by my fieldwork, as I questioned workers extensively about this issue. When asked whether they had ever accepted

[9] Remember, however, that not all of these workers are interested in gaining a permanent position.

[10] These tests do not rule out preferred clients having other characteristics—such as status—that make them attractive to contractors and make contractors willing to accept lower wages. While I cannot completely discard this hypothesis because I do not have a measure of "client status," I asked informants about this issue, and I was unable to find any evidence for it. Moreover, my examination of which clients were included in the "preferred" list suggests that high status companies are as likely to be found on the list as others.

Table 5. Career Concerns Tests

	Model 1	Model 2	Model 3	Model 4
	OLS Robust Errors by Individual		Logit Robust Errors by Individual	
	Ln (future days)	Ln (future days)	Permanent Placement	Permanent Placement
Preferred Client	.167150	—	.686463	—
	(.148072)		(.686997)	
Large Number of Contractors at Client (10th top percentile)	—	.122764	—	1.810944
		(.174489)		(1.079060)
Large Sales Volume Client (10th top percentile sales)	—	.318135	—	–1.735427
		(.207392)		(1.424934)
Ln (client size)	.012904	.011857	–.063879	–.041365
	(.015710)	(.014366)	(.100048)	(.098820)
Ln (worker's tenure)	.225297**	.231013**	–.463246*	–.365586
	(.057619)	(.057085)	(.219865)	(.247298)
Project Duration	.002647**	.002320**	.010895**	.010442**
	(.000675)	(.000679)	(.003315)	(.003810)
Years Education	–.134518	–.137843	–.174085	–.104085
	(.073590)	(.073438)	(.229733)	(.234220)
College in the Field	.201499	.209928	–.259113	–.256740
	(.167883)	(.167100)	(.712751)	(.791308)
Special Certifications	–.339492	–.345366	–.684769	–.752083
	(.201313)	(.200718)	(.868133)	(.810481)
Gender (1 = female)	–.107602	–.114187	.753034	.944574
	(.187891)	(.186730)	(.867562)	(.918366)
Age at Application	–.051131*	–.050507*	–.248785	–.269161
	(.024993)	(.024950)	(.136395)	(.151463)
Years Experience	.045422	.045023	.111743	.151213
	(.032675)	(.032702)	(.145381)	(.149773)
Previously a Contractor	.306085	.310105	–1.965605*	–1.910648*
	(.171512)	(.171519)	(.843788)	(.890584)
Year Controls	Included	Included	Included	Included
Type of Project Controls (8 dummies)	Included	Included	Included	Included
Industry Controls (13 dummies)	Included	Included	Included	Included
Constant	6.692060**	6.716166**	–13.009416**	–14.470511**
	(1.284264)	(1.277702)	(3.638885)	(3.547034)
Observations	1287	1287	1074	1074
R-squared	.41	.41	—	—

Note: Robust standard errors in parentheses. * $p < .05$; ** $p < .01$ (two-tailed tests).

lower rates than usual for projects with companies that might lead to future assignments or permanent positions, none admitted to doing so—although some claimed that, in theory, they might consider it. Both the quantitative and qualitative findings, therefore, cast doubt on the hypothesis that the brokered parties willingly and knowingly accept the broker's transfer of constraint from buyers to sellers.

DISCUSSION

A central claim in sociology is that positions in the social structure are transformed into disparities of power, influence, and economic success among actors (Marsden 1982; Simmel 1950). A great deal of progress has been made on investigating this issue by analyzing brokerage structures, as intermediaries have been shown to extract advantages from their position over and above their possession of valuable resources (Burt 1992). It is less often acknowledged, though, that if brokers can extract rents from their positions, this might have economic consequences for other parties tied to them. This article has shown this to be the case by demonstrating that a broker is able to exercise control in price setting through its transfer of the price constraints generated by buyers on to sellers. As a result, the brokered parties obtain different returns depending on the nature of the tie between the intermediary and its exchange partners on the other side of the market. Unequal economic outcomes for sellers, therefore, do not stem only from their abilities or resources, but they also arise from the relationships among other exchange partners in a given triadic network of ties. These findings provide an unusual perspective on market stratification.

I have illustrated these arguments using data from an intermediary in the high-skill IT sector, where buyers are firms and sellers are workers placed in temporary projects. The staffing industry is a typical example of a brokered market, as well as one of the most rapidly growing sectors in the U.S. economy. Although some of the dynamics observed in this setting are specific to employment intermediaries, the broad mechanisms identified here should also apply to other market brokers that match buyers and sellers (e.g., real estate agents). While there do not seem to be systematic differences between InterCo and its competitors, I do not make any claims about the representativeness of my findings. However, the in-depth information available in this setting allows us to observe some processes, mechanisms, and outcomes that are normally invisible to students of brokerage and its market consequences. Thus, the theoretical significance of this case is that it provides a window through which to view the functioning of a set of processes that are normally hidden. To my knowledge, no staffing agency so far has allowed the collection of systematic data on both prices and wages. In fact, how much these intermediaries charge clients versus how much they pay workers is one of the labor market's best kept secrets. In terms of the results themselves, it is important to point out that the broker's ability to exert control in price setting is limited by the fact that, while "voice" may not exist as such in this labor market, "exit" is certainly a very legitimate option (see Freeman 1980; Hirschman 1970). All the results are limited to those who decided to remain affiliated to this staffing agency for a certain period of time.

The findings show that InterCo has the ability to lower contractors' wages when they are matched to preferred clients. One question that logically arises is: why does the broker not pay lower wages to all employees all the time? By not doing so, this firm seems to be "leaving some money on the table"—which would be at odds with perfect rationality assumptions about profit maximization. The lack of appropriate data does not allow me to provide a conclusive answer to this question. However, my fieldwork suggests that there are both external (market-based) and internal (organizational practices) factors that might constrain the broker's ability to exercise full discretion in price setting. External constraints arise from the fact that InterCo operates in a competitive market, and its ties to both buyers and sellers need to be balanced. Prices and wages are estimated within a range of possible values determined by the external market in which the intermediary operates. A pure product market strategy would imply that low prices were charged to all clients and low wages paid to all contractors, which would damage InterCo's reputation in the labor market and impede its ability to attract high quality workers. Similarly, a pure labor market strategy would involve charging high prices to clients and thus paying high wages to all con-

tractors, which would damage its competitive position with employers. In a sense, the broker needs to balance its long-term relationships with both actors, as the external market sets (unobserved) upper and lower limits for wages. Workers need to be paid an amount similar to their reservation wage (also unobserved) over a stream of projects so they keep transacting with the broker. Lowering wages below market rates across the board would make this unfeasible.

The second reason why InterCo does not pay lower wages to all contractors all of the time has to do with established organizational practices and routines. If InterCo were behaving as a pure rational profit maximizer in the economics textbook tradition, one might expect a pricing model in which a fixed optimal pay rate (equal to each individual's minimum reservation wage) was paid to the worker (averaging across projects). In contrast, this firm seems to be following some kind of implicit rule to maintain or increase margins per transaction, and doing so by paying a variable wage to the worker—depending on the bill rate charged to the client. This method dramatically reduces the amount of information processing required to set wages. To set an optimal wage, the broker would need to aggregate data on all transactions, with the goal of calculating the minimum "reservation price" that the average seller would accept in order not to abandon the relationship. This, in turn, would require some estimation of the expected duration of ties with workers (which is very complicated in this context because individuals can sever the relationship at any time). My fieldwork indicates that placement agents, in their daily operations, do not follow this maximizing behavior to calculate optimal pay rates. This is consistent with an understanding of organizational action as "boundedly rational" (March 1994; Simon 1957), with actors routinely using heuristic devices instead of complex calculations to make everyday decisions (see also Nee and Ingram 1998). In this setting, it would be difficult for managers to think about profits ex-ante, while it is easier to aim at maintaining margins case by case. A consequence of this behavior is that the risk is pushed on to workers, whose wages then depend on the specific rates charged to the clients per transaction. These results highlight how less-than-perfect rationality in a firm's actions is somehow "enabled" by its ability to exercise control over third-party (the workers') outcomes. This is an intriguing issue that is worthy of further research.

Because I held constant the number of alternative ties for buyers and sellers, the results should not be interpreted as evidence of the broker's absolute level of market power. Rather, they indicate that, within the context of an established system of relationships, the intermediary has the potential to exercise a certain degree of control in price setting. The brokered parties in this study (sellers) are highly skilled workers. This is important, as they have other alternatives to transacting with InterCo. If as a group they had no credible alternative, the evidence of the intermediary's control would be less surprising. The client companies also have alternatives, as the broker is not a monopolist in the local labor market. These characteristics of the empirical setting allow us to focus on the relative value of resources exchanged through the ties. I have analyzed the "constraint transfer" process from buyers to sellers, although one could expect that similar processes take place from the other side of the market. Some of the results provide illustrative evidence that this is the case. However, at a general theoretical level, symmetry in this process might be dependent on the relative influence of each set of ties on the broker's business. For example, if clients are on the whole more influential for brokers than are workers (as seems to be the case in this setting), it could be expected that the effects of client-broker ties on prices would also be more significant. The extent to which this is the case, and how it differs from other contexts in which buyers and sellers are equally influential for the broker, are open empirical questions.

This article expands the literature on brokerage and market intermediation in a number of ways. First, it contributes to our knowledge about how control is exercised in brokerage by analyzing how intermediation is implemented through the broker's ability to set prices for the other two parties. There are some qualitative studies of specific brokers that analyze their exercise of agency and control generally by looking at the process through which they became successful (Caro 1982; Granovetter and McGuire 1998). Empirical quantitative analyses of how brokers exercise control in markets, however, are scarce (see Burt 2000). Second, this

article examines how intermediaries deal with the basic tensions implicit in their brokerage position by balancing competing demands through the pricing process. In particular, my results suggest that the way prices are set is closely related to the relative importance to the broker of each tie at a given time.

Another key contribution of this article is its emphasis on the consequences of intermediation for the brokered parties. Most literature on brokerage and intermediation focuses exclusively on the advantages for the intermediary (Khurana 2002). It can reasonably be expected, however, that if brokers are not just creating value through matching but are also extracting rents from their activities, this must have some consequences for those being "brokered" (see Bielby and Bielby 1999). Although this is a logical implication of brokerage theories, there is little empirical evidence in this direction. The novel mix of quantitative and qualitative data used in this article offers a nuanced perspective on this issue. On the one side, the statistical analysis allows us to observe patterns (the discount transfer) that would not be visible otherwise, as participants in this market do not talk about them openly. On the other side, the fieldwork provides a detailed account of the circumstances that allow the broker to exercise this kind of control in price setting. In particular, I show how the opaqueness of the processes by which prices are set, as well as the stickiness of relationships between exchange partners, create social frictions that shelter the broker-seller relationships from perfect competition. Social frictions created by a broker might be slightly different in other market settings but, nonetheless, one should expect that they exist if brokerage is going to be sustained.

The issue of sustainability has so far not been prominent in brokerage research. Structural holes theory (Burt 1992), in particular, is inherently a theory of disequilibrium in which the gains might be expected to disappear over time if the parties were to transact directly with each other (see Burt 2000). Yet many market intermediaries, such as venture capitalists (Podolny 2001), real estate agents (Levitt and Syverson 2005), and headhunters (Finlay and Coverdill 2000, 2002) do sustain brokerage activities over time. In the setting analyzed in this article, there are indeed ongoing ties between the intermediary and the mediated parties. Put differently, the broker is somehow managing to make a business out of maintaining the "open triad" (Burt 1992)—the "forbidden triad" in Granovetter's (1973) terms. The processes that I observe in this article can eventually help provide some answers to this issue. However, the specific conditions allowing brokerage sustainability, and how it might or might not be eroded over time, need to be explored further.

In this article, I have characterized "constraint" as stemming from relationships with buyers who are especially important for the broker, mostly because of a certain level of dependency on these key clients (Burt 1980; Emerson 1962). However, the theoretical argument holds even if we remain fully agnostic as to the reasons why the broker decides to charge lower prices to certain clients. These reasons might vary across settings, but the constraint transfer mechanism could still take place regardless. All that is required is that the intermediary has a nonneutral relationship with one side of the market, whatever the source of that nonneutrality (power, dependency, friendship, etc.). For example, studies of pricing that use embeddedness theory (Uzzi 1999; Uzzi and Lancaster 2004) show that firms may charge lower prices to certain clients. The mechanisms invoked to make this argument focus on the development of a shared logic of exchange that allows for better transmission of information and the creation of informal arrangements. Ultimately, identifying these mechanisms requires a profound knowledge of the context and adequate measures of the nature of ties in the market of interest. One could hypothesize, however, that an intermediary who is bound by embedded ties to certain buyers or sellers—and therefore offers them better prices—might still be able to transfer these short-term price constraints on to the other side of the market. If this were the case, it would highlight a rarely emphasized implication of embedded ties: that an organization's close relationships with some actors might have unexpected consequences for a third actor with whom it also has ties.

I have argued that my results on differential wages for workers depending on the broker's relationship with clients represent a puzzle from the perspective of the standard economic theory of competitive labor markets. A purely neoclassical economic argument might still be compatible with lower prices being paid to sell-

ers matched to buyers that receive discounts, but it would assume that those sellers offer a slightly lower quality product (that is, they have different resource endowments). Showing that same-quality sellers (workers) obtain different prices challenges this assumption.[11] The findings thus contribute to economic sociology models of wages by highlighting the social structure in which price setting decisions occur. In particular, they show that workers' outcomes are not determined just by their individual characteristics, nor even by their relationships with the wage setter (the intermediary), but they are also influenced by the relationships between the broker and its own clients. This suggests new empirical correlations and potential causal mechanisms that are not usually considered by standard economic studies of wages (Gibbons 2005). Finally, by ruling out that the results are the outcome of individual preferences, this article emphasizes social constraint and power dynamics rather than voluntarism and individual choice, which connects this study with traditional sociological analysis of the economy (see Baron and Hannan 1994).

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[11] This is not to say that one could not potentially build a game-theory model ex-post to explain the findings by incorporating the role of "repeated interactions," information asymmetry, and the stickiness of economic transactions. Even if one did so, a differential contribution of this article is to show how these dynamics play out—both quantitatively and qualitatively. Also, by looking at the specific social mechanisms that allow for the constraint transfer to occur, we give tangible social meaning to what in an abstract model would be identified merely as "information asymmetry."

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